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Refer : http://www.crazyforcode.com/amazon-interview-puzzle/

Amazon interview puzzle - Find Age Of Children

A man walks into a bar, orders a drink, and starts chatting with the bartender. After a while, he learns that the bartender has three children. "How old are your children?" he asks. "Well," replies the bartender, "the product of their ages is 72." The man thinks for a moment and then says, "that's not enough information." "All right," continues the bartender, "if you go outside and look at the building number posted over the door to the bar, you'll see the sum of the ages." The man steps outside, and after a few moments he reenters and declares, "Still not enough!" The bartender smiles and says, "My youngest just loves strawberry ice cream." How old are the children? A variant of the problem is for the sum of the ages to be 13 and the product of the ages to be the number posted over the door. In this case, it is the oldest that loves ice cream. Then how old are they?

Answer :

First, determine all the ways that three ages can multiply together to get 72:

72 1 1 (quite a feat for the bartender)
36 2 1
24 3 1
18 4 1
18 2 2
12 6 1
12 3 2
9 4 2
9 8 1
6 3 3

6 6 2

6 4 3

As the man says, that's not enough information; there are many possibilities. So the bartender tells him where to find the sum of the ages--the man now knows the sum even though we don't. Yet he still insists that there isn't enough info. This must mean that there are two permutations with the same sum; otherwise the man could have easily deduced the ages.

The only pair of permutations with the same sum are 8 3 3 and 6 6 2, which both add up to 14 (the bar's address). Now the bartender mentions his "youngest"--telling us that there is one child who is younger than the other two. This is impossible with 8 3 3--there are two 3 year olds. Therefore the ages of the children are 6, 6, and 2.

Pedants have objected that the problem is insoluble because there could be a youngest between two three year olds (even twins are not born exactly at the same time). However, the word "age" is frequently used to denote the number of years since birth. For example, I am the same age as my wife, even though technically she is a few months older than I am. And using the word "youngest" to mean "of lesser age" is also in keeping with common parlance. So I think the solution is fine as stated.

In the sum-13 variant, the possibilities are:

11 1 1

10 2 1

9 3 1

9 2 2

8 4 1

8 3 2

7 5 1

7 4 2

7 3 3

6 6 1

6 5 2

6 4 3

The two that remain are 9 2 2 and 6 6 1 (both products equal 36). The final bit of info (oldest child) indicates that there is only one child with the highest age. This cancels out the 6 6 1 combination, leaving the childern with ages of 9, 2, and 2.

Red and Blue Balls in a Bag

November 4, 2013 1:32 pm 5 Comments crazyadmin

Problem:

You have 20 Blue balls and 10 Red balls in a bag. You put your hand in the bag and take off two at a time. If they're of the same color, you add a Blue ball to the bag. If they're of different colors, you add a Red ball to the bag. What will be the color of the last ball left in the bag?

Note: Assume you have a big supply of Blue and Red balls for this purpose. When you take the two balls out, you don't put them back in, so the number of balls in the bag keeps decreasing.

Once you tackle that, what if there are 20 blue balls and 11 red balls to start with?

Solution:

There can be 3 possible cases taking off 2 balls from bag.
a) If we take off 1 Red and 1 Blue, in fact we will take off 1 Blue
b) If we take off 2 Red, in fact we will take off 2 Red (and add 1 Blue)
c) If we take off 2 Blue, in fact we will take off 1 Blue

So In case of (a) or (c), we are only take off one Blue ball. Also, we always take off Red balls two by two.

1) 20 Blue, 10 Red balls
If there are 10 (even) number of Red balls, we can not have one single Red ball left in the bag, so the last ball will be Blue.

2) 20 Blue, 11 Red balls
Now as the no. of Red balls is odd, there will be one single Red ball in the bag with other Blue balls, and whenever we remove 1 Red and 1 Blue ball, we end up taking off only the Blue ball. So the Red ball will be the last ball in the bag.

Heaven or Hell Puzzle

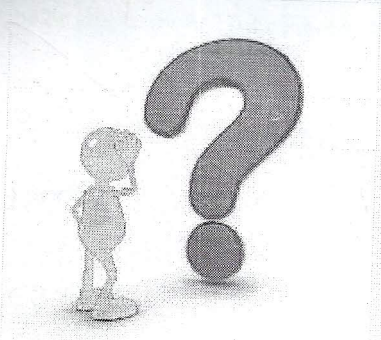
October 24, 2013 8:10 am 34 Comments crazyadmin

Puzzle:

You are standing before two doors. One of the path leads to heaven and the other one leads to hell. There are two guardians, one by each door. You know one of them always tells the truth and the other always lies, but you don't know who is the honest one and who is the liar.

You can only ask one question to one of them in order to find the way to heaven. What is the question?

This is one of the classic puzzles asked in Infosys interview.



Puzzle Solution:

The question you should ask is "If I ask the other guard about which side leads to heaven, what would he answer?". It should be fairly easy to see that irrespective of whom do you ask this question, you will always get an answer which leads to hell. So you can chose the other path to continue your journey to heaven.

This idea was famously used in the 1986 film Labyrinth.

Here is the explanation if it is yet not clear.

Let us assume that the left door leads to heaven.

If you ask the guard which speaks truth about which path leads to heaven, as he speaks always the truth, he would say "left". Now that the liar , when he is asked what "the other guard (truth teller) " would answer, he would definitely say "right".

Similarly, if you ask the liar about which path leads to heaven, he would say "right". As the truth teller speaks nothing but the truth, he would say "right" when he is asked what "the other guard(liar) " would answer. So in any case, you would end up having the path to hell as an answer. So you can chose the other path as a way to heaven.

King and Wine Bottles

September 5, 2013 6:40 pm | 19 Comments | crazyadmin

Problem:

A bad king has a cellar of 1000 bottles of delightful and very expensive wine. A neighboring queen plots to kill the bad king and sends a servant to poison the wine. Fortunately (or say unfortunately) the bad king's guards catch the servant after he has only poisoned one bottle. Alas, the guards don't know which bottle but know that the poison is so strong that even if diluted 100,000 times it would still kill the king. Furthermore, it takes one month to have an effect. The bad king decides he will get some of the prisoners in his vast dungeons to drink the wine. Being a clever bad king he knows he needs to murder no more than

10 prisoners – believing he can fob off such a low death rate – and will still be able to drink the rest of the wine (999 bottles) at his anniversary party in 5 weeks time. Explain what is in mind of the king, how will he be able to do so ? (of course he has less then 1000 prisoners in his prisons)



Solution:

Think in terms of binary numbers. (now don't read the solution, give a try).

Number the bottles 1 to 1000 and write the number in binary format.

bottle 1 = 0000000001 (10 digit binary)

bottle 2 = 0000000010

bottle 500 = 0111110100

bottle 1000 = 1111101000

Now take 10 prisoners and number them 1 to 10, now let prisoner 1 take a sip from every bottle that has a 1 in its least significant bit. Let prisoner 10 take a sip from every bottle with a 1 in its most significant bit. etc.

prisoner = 10 9 8 7 6 5 4 3 2 1

bottle 924 = 1 1 1 0 0 1 1 1 0 0

For instance, bottle no. 924 would be sipped by 10,9,8,5,4 and 3. That way if bottle no. 924 was the poisoned one, only those prisoners would die.

After four weeks, line the prisoners up in their bit order and read each living prisoner as a 0 bit and each dead prisoner as a 1 bit. The number that you get is the bottle of wine that was poisoned.

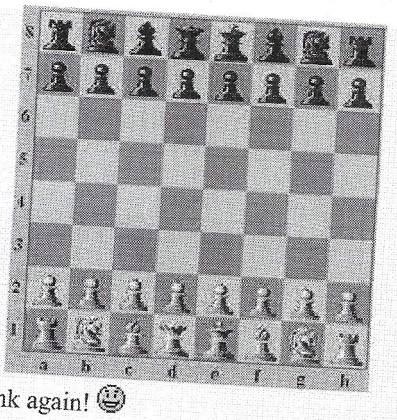
1000 is less than 1024 (2^10). If there were 1024 or more bottles of wine it would take more than 10 prisoners.

Squares on a chess board?

July 30, 2013 7:33 pm | 3 Comments | crazyadmin

Problem:

How many squares are on a chess board?



If you thought the answer is 64, think again! 😃

How about all the squares that are formed by combining smaller squares on the chess board (2×2, 3×3, 4×4 squares and so on)?

A 1×1 square can be placed on the chess board in 8 horizontal and 8 vertical positions, thus making a total of 8 x 8 = 64 squares. Let's consider a 2×2 square. There are 7 horizontal positions and 7 vertical positions in which a 2×2 square can be placed. Why? Because picking 2 adjacent squares from a total of 8 squares on a side can only be done in 7 ways. So we have 7 x 7 = 49 2×2 squares. Similarly, for the 3×3 squares, we have 6 x 6 = 36 possible squares. So here's a break down.

1×1 8 x 8 = 64 squares
2×2 7 x 7 = 49 squares
3×3 6 x 6 = 36 squares
4×4 5 x 5 = 25 squares
5×5 4 x 4 = 16 squares
6×6 3 x 3 = 9 squares
7×7 2 x 2 = 4 squares
8×8 1×1 = 1 square

1^2 + 2^2 + 3^2 + . . . n ^2

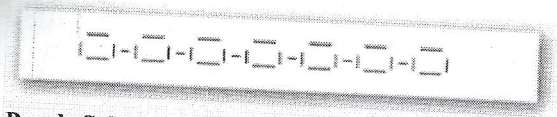
Total = 64 + 49 + 36 + 25 + 16 + 9 + 4 + 1 = **204** squares

Gold Bar Puzzle

July 25, 2013 8:54 pm | 10 Comments | crazyadmin

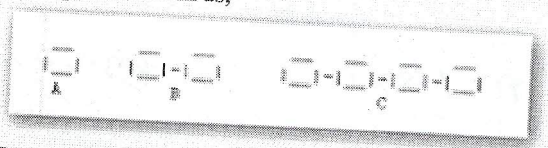
Puzzle:

You've got someone working for you for seven days and a gold bar to pay him. The gold bar is segmented into seven connected pieces. You must give them a piece of gold at the end of every day. What and where are the fewest number of cuts to the bar of gold that will allow you to pay him 1/7th each day?



Puzzle Solution:

Lets split the chain as,



Day 1: Give A (+1)
Day 2: Get back A, give B (-1, +2)
Day 3: Give A (+1)
Day 4:Get back A and B, give C (-2,-1,+4)
Day 5:Give A (+1)
Day 6:Get back A, give B (-1,+2)
Day 7:Give A (+1)

Boys and Girls

July 2, 2013 3:41 pm 5 Comments crazyadmin

Question:

In a country where everyone wants a boy, each family continues having babies till they have a boy. After some time, what is the proportion of boys to girls in the country? (Assuming probability of having a boy or a girl is the same)



Answer:

"very simple". Half the couples have boys first, and stop. The rest have a girl. Of those, half have a boy second, and so on.

So suppose there are N couples. There will be N boys.

1/2 have a boy and stop: 0 girls
1/4 have a girl, then a boy: N/4 girls
1/8 have 2 girls, then a boy: 2*N/8 girls
1/16 have 3 girls, then a boy: 3*N/16 girls
1/32 have 4 girls, then a boy: 4*N/32 girls
...